

FILTER DESIGN TEST CASE

Lattice Filter - WiFi 6E (6 GHz) for UWB (Ultra-Wideband) Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting WiFi 6E (6 GHz) for UWB (Ultra-Wideband) Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 1168.4 MHz and bandwidth of 181.4 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the UWB (Ultra-Wideband) Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: WiFi 6E (6 GHz)
- Application: UWB (Ultra-Wideband) Filter
- Substrate: 128° YX LiNbO3
- Package: SiP (System in Package)
- Design Tool: COMSOL Multiphysics 6.2
- Design Center: Cedar Rapids Lab

This specification applies to all variants of the Lattice Filter design targeting WiFi 6E (6 GHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1168.4 MHz
Bandwidth (3 dB):	181.4 MHz
Insertion Loss Target:	<= 1.2 dB
Return Loss Target:	>= 13.0 dB
Out-of-Band Rejection:	>= 31.0 dB
Isolation (Duplexer):	>= 19.0 dB
Q Factor:	4297
Temperature Coefficient:	-15.8 ppm/°C
Die Size:	2.6 x 1.6 mm
Wafer Size:	6-inch
Number of Resonators:	10
Electrode Material:	Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open COMSOL Multiphysics 6.2 and load the Lattice Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 1168.4 MHz, BW 181.4 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 1.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (SiP (System in Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Keysight N5227B PNA.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 3505 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 1.2 dB
- Return loss in passband shall be >= 13.0 dB
- Out-of-band rejection at +/- 272 MHz offset >= 31.0 dB
- Group delay variation across passband shall be <= 7 ns
- Temperature drift over -40°C to +85°C shall be <= 2.3 MHz
- Power handling shall be >= +34 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 98%

6. TEST RESULTS

Measurement Date: 2025-02-07
Devices Tested: 100
Insertion Loss (meas): 0.91 dB
Return Loss (meas): 14.9 dB
Rejection (meas): 34.0 dB
Isolation (meas): 24.1 dB
Q Factor (meas): 4297
Group Delay Variation: 1.5 ns
Power Handling: +30 dBm
Die Yield: 85.5%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated below-target performance for WiFi 6E (6 GHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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